NPLify · P0 Technical Baseline

Calculation Specification · Draft v5.9

New XP Technologies Limited

11 October 2026 · Confidential

Numbered with the ERD edition it is built against. Supersedes Draft v1.0 (ERD v4.0).

Status: Built against ERD & Data Model Draft v5.9 (decisions D1–D46) and the Ledger Posting Design v5.9. It states every calculation the engine performs as a precise formula, in the order it is performed, with the rounding rule at each step, and gives acceptance vectors of two kinds: the Ledger Posting Design's worked deals, produced by the same pricing function this document imports, and NPL's own figures from its fee table, recomputed here and compared. The engine passes when it reproduces every vector.
What changed since v1.0: fee parts are percentages of the principal at the basis rate and never compound (NPL, 11 October); a target-out quote applies the fee as a markup on the target (NPL's sheets); the rate basis is market or agent_rate with the partner's cost inside its rate or stated separately; tiers, the customer-facing figures, the rounding menu, cash rounding, same-currency pairs, the markup-share base, rebates, the four variance kinds, the forex true-up on invoice instalments, earnings retained and offset between partners, loss splits, and the reseller arithmetic are all specified (D20–D46, with the same-day amendment of D46: views over postings, the two-invoice links, FX_RESULT, the target-out rule). v5.9 regenerates this document against ERD v5.9 and the Ledger Posting Design v5.9 (the corrections of the cross-document review): no formula changes; the loss split of §16 is now posted as one LOSS_RECOVERY per share.

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1 · Conventions

- **Rates** are stored normalised to the pair's `quote_direction` (out units per 1 in unit unless the pair says in_per_out) at `CURRENCY_PAIR.rate_precision`. A partner's rate is stored as quoted (`PARTNER_RATE_VERSION.quoted_rate`, `quoted_direction`) and the normalised `rate` is derived: equal to the quote when the directions agree, 1 ÷ quote at the pair's precision when they differ (D28).
- **Amounts** are exact decimals; fiat and USDT amounts are computed to 2 decimals (BTC-like assets to 8) with round half up, then rounded as the pair's option says (§9). Every intermediate figure in this document is computed, not typed.
- **Percentages** are stored as fractions (1.4 % = 0.014). `sender_pays_pct` + `receiver_pays_pct` = `FEE_STRUCTURE.pct`.
- **Precision of a check**: an acceptance vector matches when the computed figure and NPL's figure agree to the cent, or to the unit where NPL's sheet rounds to the unit.

2 · Rate basis, derived rates and the breach check

`FEE_STRUCTURE.rate_basis` says which rate the sender is priced on: **market** (the fee is disclosed; the partner's cost comes out of NPL's fee parts — Evo, the reseller projects) or **agent_rate** (a calculated rate, breakdown not disclosed; NPL keeps the whole fee and the partner's cost sits on top — BF via Ali, D22). The partner's own pricing is `PARTNER_PAIR.partner_pricing`: **fee_in_rate** (its cost is inside its rate, never posted; Ali, assumed 1 %) or **fee_on_market** (it converts at market and states `partner_fee_pct` separately; Jeton 1.25 %, Aquanow 0.80 %).

```
basis_rate     = market_rate                          if rate_basis = market
               = partner_rate_version.rate          if rate_basis = agent_rate
rate           = quoted_rate                          if quoted_direction = pair.quote_direction
               = round(1 ÷ quoted_rate, rate_precision)   otherwise                        (D28)
spread         = 1 − partner_rate ÷ market_rate      (what the partner keeps against market,
fee_in_rate)
breach         = spread > avg_2w_spread × (1 + variance_threshold_rel)   → the pair freezes
until Finance decides
```

3 · Fee parts on the principal

The sender's part and the receiver's part are each a percentage of the **principal at the basis rate**. They are never compounded on each other (NPL, 11 October: 300,000 USDT × 0.9000 − 2 % = 264,600, not 264,622.68).

```
principal          = amount_in × basis_rate                                  (fixed_in)
fee_sender_part    = principal × sender_pays_pct
fee_receiver_part  = principal × receiver_pays_pct
sender_rate        = basis_rate × (1 − sender_pays_pct)                    (the rate the
sender is given)
gross_out          = amount_in × sender_rate              = principal − fee_sender_part
client_net         = gross_out − fee_receiver_part        = principal × (1 − sender_pays_pct −
receiver_pays_pct)
```

- 10,000 USDT × market 0.9000 = 9,000.00 EUR (the principal at the basis rate)
- sender part = 9,000 × 1.40 % = 126.00; receiver part = 9,000 × 0.60 % = 54.00
- sender rate = 0.9000 × (1 − 1.40 %) = 0.88740; gross out = 10,000 × 0.88740 = 8,874.00
- client net = 9,000 × (1 − 2.00 %) = 8,820.00 = gross out − receiver part

Confirmed rule (NPL, 11 October). Both parts are on the principal. NPL's fee sheet for Evo prints the receiver side as "EUR 25,704 − 0.6 % = 25,550", which takes the receiver's part from the gross out; on the principal it is 25,547.88, a difference of 0.6 % × 1.4 % of the principal (about 0.008 %). The sheet's receiver line is to be brought in line with the rule; the engine follows the rule.

4 · Fixed and variable fee modes; the minimum-margin check

In **fixed** mode `pct` is applied as above. In **variable** mode the fee is whatever leaves NPL its minimum margin after the partner's cost, within a floor and a cap: this is the check that decides whether 2 % is enough when the agent rate is high.

```
required_pct   = min_margin_pct + spread            if rate_basis = market   (the partner's
cost comes out of the fee)
               = min_margin_pct                     if rate_basis = agent_rate
fee_pct        = min(max_markup_pct, max(min_markup_pct, required_pct))
below_margin   = fee_pct < required_pct             → warn, never refuse a quote; Finance may
```

```
ask the partner for a better rate or
                                                    recommend a fee above the cap
(raise_fee_cap, Management approves, D41)
```

5 · Fee tiers by monthly volume (D21)

```
tier           = the FEE_TIER of the structure with the highest min_monthly_volume ≤
PROJECT_MONTH_VOLUME at quote
portion_1      = max(0, threshold_of_next_tier − PROJECT_MONTH_VOLUME)   capped at amount_in,
priced at tier.pct
portion_2      = amount_in − portion_1                                    priced at
next_tier.pct
fee_parts      = Σ portions × basis_rate × pct (sender and receiver parts in the structure's
proportion)   — the blended sum
```

GDC: 2.5 % up to USD 500,000 a month, 2.25 % above; a leg that takes the month past the threshold is split into two DEAL_FEE_TIER portions.

6 · Amount modes: fixed_in and target_out

A collect-first deal prices what arrived (**fixed_in**, §3). A quote-first deal for a given out amount (**target_out**: an invoice, a GDC USD amount, a reseller's EUR) applies the fee as a markup on the target, which is how NPL's sheets print it (EUR 50,000 × 1.01 ÷ rate; USD 84,482 + 2.5 %).

```
amount_in      = target_out × (1 + sender_pays_pct) ÷ basis_rate          (then cash-rounded if
collected in cash, §9)
sender_rate    = basis_rate ÷ (1 + sender_pays_pct)                       (printed as target ÷
amount_in)
client_net     = target_out × (1 − receiver_pays_pct) + cash_rounding_surplus
fee_sender_part = amount_in × basis_rate − (target_out + cash_rounding_surplus)
```

Note: × (1 + s) and ÷ (1 − s) differ by s^2 (0.01 % at a 1 % fee). NPL's convention, × (1 + s), is adopted; the fixed_in formula keeps × (1 − s). The cash-rounding surplus is the client's (in NPL-GR, NPL-GR's).

- NPL-GR: invoice 9,450 EUR ÷ 0.8708 = 10,852.09 USDT to deliver
- amount_in (exact) = 10,852.09 × (1 + 2 %) × 32.50 THB per USDT = 359,746.78 THB; collected as cash, rounded up to the 1,000 = 360,000 THB
- client net = 360,000 ÷ 32.50 × (1 − 2 %)… no: in target mode the fee is on the target, so client net = target + the cash-rounding surplus = 10,859.73 USDT (7.63 surplus); sender part = amount_in × market − client net = 217.19
- Ali converts at 1 ÷ 32.83: delivered 10,966.15; NPL earnings = 10,966.15 − 10,859.73 = 106.42 = the 2 % markup less Ali's 1 %

7 · Customer-facing figures (D22)

```
customer_facing_fee_pct  = sender_pays_pct + receiver_pays_pct        if
customer_facing_fee_basis = actual
                          = sender_pays_pct on the agent rate          if
sender_pays_on_agent_rate
                          = FEE_STRUCTURE.customer_facing_fee_pct      if fixed   (BF via Ali
prints 1 %)
customer_facing_rate     = sender_rate ÷ (1 − customer_facing_fee_pct)   — the "market rate"
the quote prints, derived, never typed
```

8 · What the partner delivers, and NPL's earnings

```
fee_in_rate:    delivered = amount_in × partner_rate;          partner_cost (inferred) =
amount_in × (market − partner_rate), information only
fee_on_market:  converted = amount_in × market_rate;  partner_cost = converted ×
partner_fee_pct;  delivered = converted − partner_cost
earnings        = delivered − client_net                      (NPL's margin: EARN_GROSS −
EXP_PARTNER)
EARN_GROSS      = converted − client_net                       (fee_on_market: the full fee
parts, the partner's cost booked apart as EXP_PARTNER;
                                                                fee_in_rate: the fee parts less
the partner's hidden cost − equal to earnings)
retained_in     = earnings_in_out_currency ÷ partner_rate      when the margin is kept in the
collection currency (CONVERSION.residual_in, D46)
converted_in    = amount_in − retained_in − offset_in          (what the partner actually
converts, §13)
```

- Ali (fee_in_rate) converts at 0.8910: delivered = 10,000 × 0.8910 = 8,910.00; NPL earnings = 8,910.00 − 8,820.00 = 90.00; Ali's inferred cost = 10,000 × (0.9000 − 0.8910) = 90.00, information only
- Jeton (fee_on_market 1.25 %) converts at market: 9,000.00; partner cost = 9,000.00 × 1.25 % = 112.50; delivered 8,887.50; fee charged 180.00 (EARN_GROSS); NPL earnings = 8,887.50 − 8,820.00 = 67.50

9 · Rounding and precision (D25, D46)

```
payout:   amount_rounding = exact | truncate_unit | nearest_unit | round_up_unit | nearest_n |
round_up_n (rounding_unit)
          payable  = round(client_net, option);   residual = client_net − payable   (signed)
          residual > 0 → stays client money in DUE;  residual < 0 → NPL's cost, EXP_ROUNDING,
out of its margin
cash in:  cash_rounding = truncate_n | nearest_n | round_up_n (cash_rounding_unit: AED 100, USD
cash 10, THB 1,000)   − the collection side;
          a cash payout rounds by the pair's amount_rounding (round_up_n 1,000 for THB hand-
overs)
          amount_in = round(exact amount_in, option);   surplus = (amount_in − exact) ×
basis_rate × (1 − fee)   → the client's
rates:    rate_precision per pair (6 for EUR/USDT, 8 for BTC);   intermediate amounts 2 dp (8
for BTC), round half up
```

Bank transfers go out exact, cents included; rounding to a unit applies where the pair says so, in practice to cash hand-overs (NPL, 11 October). USDT paid by reseller customers is rounded up.

- THB payout in cash: client net 95,844.00 THB, cash_rounding round_up_n 1,000 → 96,000.00; residual = 95,844.00 − 96,000.00 = -156.00 (negative: NPL's cost, EXP_ROUNDING); NPL's pool gets 966.00 -156.00 = 810.00

10 · Same-currency pairs (D33)

```
basis_rate = partner_rate = 1;   fee parts and client_net as in §3;   earnings =
fee_sender_part + fee_receiver_part
```

Evo EUR → EUR: 2.8 %, sender 1.4 % and receiver 1.4 %: 5,000 EUR → client net 4,860.00, NPL 140.00.

11 · Markup shares and rebates (D26, D27, D46)

```
share base (share_of_net_markup) = fee_sender_part + fee_receiver_part − partner_cost
(rebates never in the base)
```

```
share_i        = base × pct_i;   Σ pct_i ≤ 100 %;   accrued and posted with the conversion
  npl_pays:      SHARE_PAYABLE to the party, off EARN_GROSS
  party_retains: the party holds the funds; NPL's remainder = base − Σ shares →
SHARE_RECEIVABLE
share_of_earnings: share = earnings × pct;   fixed_pct: priced into the sender rate
rebate         = amount_in × pct                (basis pct_of_amount, the only basis; Jeton
0.4 %, Aquanow 0.3 %, Novi 0.3 %), in PARTNER_REBATE_RULE.payout_currency
               accrued with the conversion → REBATE_RECEIVABLE / EARN_REBATE; the statement
clears it, a difference trues up EARN_REBATE
```

- Raeen sub-account at Aquanow: 10,000 × 0.9000 = 9,000; sender pays 4 % = 360.00; client net 8,640.00; Aquanow 0.80 % of 9,000 = 72.00; delivered 8,928.00
- share base = fee parts 360.00 − partner cost 72.00 = 288.00; Raeen 50 % = 144.00; NPL remainder 144.00; the 0.3 % rebate (30.00 USDT) is outside the base

12 · Variances, by cause (D29, D45, D46)

```
expected_out   = converted_in × partner_rate_version.rate           (as priced)
actual_out     = what the partner booked (partner_calculates) or expected_out (npl_calculates)
variance       = expected_out − actual_out                          (+ = NPL received less than
priced)
client_net     unchanged (write-once);  pool = actual_out − client_net;  the variance posts to
  VAR_CONVERSION   partner_calculates differences and basis variance on a reroute
  VAR_CUTOFF       rate stamped at lock vs rate used after the partner's cutoff
(conversion_due_date)
  VAR_RATE_HONOUR  an expired quote's rate honoured by override vs the day's rate
  VAR_FX_TIMING    an obligation priced on one day and discharged on another (§14)
```

- partner_calculates: 3,000 USDT expected at 0.8910 = 2,673.00, booked at 0.8905 = 2,671.50; variance = 1.50 → VAR_CONVERSION; client net 2,646.00 unchanged
- cutoff timing: locked 0.8910, converted next day at 0.8890 → 6.00 → VAR_CUTOFF
- honoured expired rate: 0.8910 honoured, Ali at 0.8860 → 15.00 → VAR_RATE_HONOUR

13 · Earnings retained in the collection currency and offset between partners (D46)

```
retained_in    = earnings ÷ partner_rate                          (this deal's margin kept
unconverted, residual_in)
offset_to      = from_amount ÷ to_partner_rate_of_day          (EARNINGS_OFFSET: NPL's margin
at the from-partner, in USDT at the to-partner)
converted_in   = amount_in − retained_in − offset_to
client_out     = converted_in × partner_rate                     = client_net − from_amount
already delivered at the from-partner
neutral for the client: Σ client entitlement unchanged; NPL's EARN_GROSS for the deal is
recognised in the collection currency
```

- NPL's margin at Jeton (EUR) 2,025.00 ÷ Ali's rate of the day 0.8910 = 2,272.73 USDT taken from the client's unconverted balance at Ali
- Ali deal margin 900.00 EUR ÷ 0.8910 = 1,010.10 USDT retained (residual_in)
- Ali converts 100,000 − 2,272.73 − 1,010.10 = 96,717.17 USDT × 0.8910 = 86,175.00 EUR = client net 88,200.00 − 2,025.00 already delivered at Jeton

14 · Invoices paid in instalments and the forex true-up (D24, D46)

```
reserved_i     = obligation_discharged_i × DEAL_GROUP.invoice_rate   (payout currency per
invoice currency at pricing: the stored base)
paid_i         = obligation_discharged_i × SETTLEMENT_LINE.discharge_rate_i   (same direction,
at payment)
true_up_i      = reserved_i − paid_i                                 (+ gain, − loss) →
VAR_FX_TIMING against NPL's pool
INVOICE_BALANCE = vendor invoice: amount − Σ obligation_discharged over released lines not
returned;  resale invoice: amount − Σ DEAL_GROUP.invoice_amount collected
```

- invoice_rate = 1.148369 USDT per EUR (0.8708 the other way); instalment 1: 3,000 EUR × 1.148369 = 3,445.11 USDT paid at the same rate it was reserved at; true-up 0
- instalment 2: discharge_rate 1.142505; paid 6,000 × 1.142505 = 6,855.03; reserved 6,000 × 1.148369 = 6,890.22; true-up = 35.19 USDT gain → VAR_FX_TIMING (a debit when the rate moves against NPL)
- margin line: 450 EUR × 1.142505 = 514.13; reserved 450 × 1.148369 = 516.77; true-up 2.64

15 · The reseller project (NPL-GR)

```
reseller_margin   = resale_invoice − its vendor_invoice (INVOICE.vendor_invoice_id)   (EUR;
fixed when the two invoices are posted)
target_out        = resale_invoice ÷ pricing_rate                          (USDT to deliver)
amount_in         = target_out × (1 + remittance_markup) ÷ market_rate     (§6; cash-rounded
if THB cash)
remittance_margin = earnings (§8) + Σ true_up_i (§14) − fees               (NPL's example: 2
% − 1 % Ali − 0.1 % forex = 0.9 %…1.2 %)
NPL_PROFIT        = Σ remittance_margin + Σ reseller_margin                (per currency; one
figure through REPORTING_VALUE)
```

16 · Pools, squaring, loss splits, balance conversion, returns, network fees

```
squaring       : pool_ccy < 0 after an absorbed fee → next conversion includes (−pool_ccy ÷
partner_rate) of NPL's own USDT
loss split     : sender_share + npl_share + partner_share + client_share = LOSS_EVENT.amount ≤
amount verified or released;  one LOSS_RECOVERY posting per share (T41a–c)
balance conv.  : out = amount × partner_rate (no fee unless a FEE_DECISION applies §3);
rounding as the pair says
return leg     : out = returned_amount × partner_rate; markup only by FEE_DECISION; the first
conversion's margin stays earned
bank fee       : treatment waived (client bears) | carry_forward (SHORTFALL, topped up) |
absorbed or covered (NPL, out of the pool)
network fee    : USDT; zero and unposted when none applies; absorbed by NPL or charged back to
the sender per project policy
```

- bounce fee 15.00 GBP absorbed with no GBP pool at Ali → pool −15.00; squared at the next conversion: 15.00 ÷ 0.7722 = 19.43 USDT of NPL's own converted alongside the client's
- loss 10,000.00 USDT → EXP_LOSS; split partner 4,000.00 + sender 2,000.00 + client 1,000.00 + NPL 3,000.00 = 10,000.00; EXP_LOSS after recoveries = 3,000.00

17 · Reporting values (D43)

```
REPORTING_VALUE(figure, ccy, as_of) = figure × latest market_rate(platform pair ccy →
reporting_currency), with the rate, its time and a stale flag
never posted, quoted or settled; the only control reading it is a ceiling stated in the
reporting currency
```

18 · Acceptance vectors A — the ledger's worked deals

Produced by the pricing function the Ledger Posting Design v5.9 imports; the posting tables there show the same figures line by line. Rates: market 0.9000 EUR per USDT, Ali 0.8910; Evo sender 1.4 %, receiver 0.6 % (L3: 1.4 / 1.4; L6: 4 % / 0; L7: 2 % markup on the target, receiver 0).

#	Case	amount_in	sender rate	gross out	sender part	receiver part	client net	delivered	partner cost	NPL earnings	payable	residual
L1	Deal 1 — Evo, 10,000 USDT → EUR at Ali (fee_in_rate), market 0.9000, Ali 0.8910, sender 1.4 % + receiver 0.6 % on market	10,000.00 USDT	0.887400	8,874.00	126.00	54.00	8,820.00	8,910.00	0.00	90.00	8,820.00	0.00
L2	Deal 3 — 10,000 USDT → EUR at Jeton (fee_on_market 1.25 %), same fee	10,000.00 USDT	0.887400	8,874.00	126.00	54.00	8,820.00	8,887.50	112.50	67.50	8,820.00	0.00
L3	Same-currency 5,000 EUR → EUR at rate 1, Evo EUR → EUR 2.8 % split 1.4 / 1.4	5,000.00 EUR	0.986000	4,930.00	70.00	70.00	4,860.00	5,000.00	0.00	140.00	4,860.00	0.00
L4	3,000 USDT → THB cash at Ali (market 32.60, Ali 32.27), round_up_n 1,000	3,000.00 USDT	32.143600	96,430.80	1,369.20	586.80	95,844.00	96,810.00	0.00	966.00	96,000.00	-156.00
L5	2,000 USDT → GBP at Ali (market 0.7800, Ali 0.7722)	2,000.00 USDT	0.769080	1,538.16	21.84	9.36	1,528.80	1,544.40	0.00	15.60	1,528.80	0.00
L6	Raeen sub-account: 10,000 USDT → EUR at Aquanow (fee_on_market 0.80 %), sender 4 % on market, receiver 0	10,000.00 USDT	0.864000	8,640.00	360.00	0.00	8,640.00	8,928.00	72.00	288.00	8,640.00	0.00
L7	NPL-GR: target 10,852.09 USDT (= 9,450 EUR at 0.8708) collected in THB cash at Ali, 2 % markup on the target, cash round-up 1,000; Ali 1 ÷ 32.83 vs market 1 ÷ 32.50	360,000.00 THB	0.030166	10,859.73	217.19	0.00	10,859.73	10,966.15	0.00	106.42	10,859.73	0.00
L8	Pattern N: 300,000 USDT → EUR at Jeton (1.25 %)	300,000.00 USDT	0.887400	266,220.00	3,780.00	1,620.00	264,600.00	266,625.00	3,375.00	2,025.00	264,600.00	0.00
L9	Pattern N: 100,000 USDT → EUR at Ali	100,000.00 USDT	0.887400	88,740.00	1,260.00	540.00	88,200.00	89,100.00	0.00	900.00	88,200.00	0.00

L4's payable is the cash hand-over rounded up to 1,000 THB (residual negative, NPL's cost). L7's client net includes the 7.63 USDT cash-rounding surplus. Variances, offsets and the forex true-ups are in §12–§14 with their figures.

19 · Acceptance vectors B — NPL's own figures, recomputed

Each row takes the inputs NPL's fee sheet shows, applies this document's formulas, and compares with the figure the sheet prints. "matches" means to the cent, or to the unit where the sheet rounds.

Case	Formula applied	Computed	NPL's sheet	Result	Note
Evo sender: 29,325 USDT at market 0.888977323, sender pays 1.40 %	fixed_in; sender rate = market × (1 − 1.40 %) = 0.876531640478	25,704	25,704	**matches**	NPL's sheet prints the sender rate 0.876531640478 and EUR 25,704 (rounded to the unit)
Evo receiver: the same deal, receiver pays 0.60 %	fee parts on the principal: 29,325 × 0.888977323 × (1 − 1.40 % − 0.60 %)	25,548	25,550	**sheet differs**	NPL's sheet applies 0.60 % to the EUR 25,704 (25,550); on the principal, the confirmed rule, it is 25,547.88 — the sheet's line is the one to correct (see §3)
Evo selected senders: EUR 50,000 target, market 1 ÷ 0.8652 USDT per EUR, sender pays 1.00 %	target_out; amount_in = target × (1 + 1.00 %) × market in_per_out = 50,000 × 1.01 × 1.155802126676	58,368	58,368	**matches**	NPL prints the final rate 1.16736014794267 = market × 1.01
BF-Raeen via Jeton: 200,000 USDT at 0.8610999, fee 1 %	fixed_in; 200,000 × 0.8610999 × (1 − 1 %)	170,498	170,498	**matches**	
BF-Raeen via Ali: 300,000 USDT at 0.864189, fee 1 %	fixed_in; 300,000 × 0.864189 × (1 − 1 %)	256,664	256,664	**matches**	the printed rate is recomputed so the sender-facing fee reads 1 % (D22)
Raeen sub-account: 50,000 USDT at 0.8650036, fee 4 %	fixed_in; 50,000 × 0.8650036 × (1 − 4 %)	41,520.17	41,520.17	**matches**	
GDC: USD 84,482 target, fee 2.5 %, converted at a rate of 1 USDT per USD	target_out; 84,482 × (1 + 2.5 %) ÷ 1.0000	86,594	86,594	**matches**	the sheet's figure implies a rate of exactly 1
GDC above USD 500,000 a month: tier 2.25 %	target_out; 84,482 × (1 + 2.25 %)	86,383	86,383	**matches**	FEE_TIER chosen from PROJECT_MONTH_VOLUME (D21)
27 Group (reseller): THB 17,000,000 at 0.0259463782412186 EUR per THB, fee 2 %	fixed_in; sender rate = 0.0259463782412186 × 0.98 = 0.02542745; 17,000,000 × 0.02542745	432,266.66	432,266.66	**matches**	NPL rounds the EUR to the unit: 432,267
27 Group: the same THB at Ali, agent rate 1 ÷ 33.83 USDT per THB	what the partner delivers: 17,000,000 ÷ 33.83	502,513	502,513	**matches**	agent fee 0.64 % inside the rate against market 1 ÷ 33.613
27 Group balance clearance: EUR 329,408 target, market 38.13570206 THB per EUR, fee 2 %	target_out; 329,408 × 1.02 × 38.13570206	12,813,449.45	12,813,449.45	**matches**	then cash-rounded by NPL to THB 12,813,000 (nearest 1,000; the pair's cash_rounding)
Bala (reseller): EUR 2,220 target, market 1 ÷ 0.8716, fee 2 %	target_out; 2,220 × 1.02 × 1.147315282240	2,597.98	2,597.98	**matches**	then rounded up to USDT 2,600 (D25: USDT paid by reseller customers rounded up)
66 Group (reseller): EUR 239,149 target, market 1 ÷ 0.8796, fee 1.5 %	target_out; 239,149 × 1.015 × 1.136880400182	275,962.07	275,962.07	**matches**	rounded to USDT 275,962
TripleSeven/GZ (reseller): EUR 4,097 target, market 1 ÷ 0.8598, fee 1.5 %	target_out; 4,097 × 1.015 × 1.163061177018	4,837	4,837	**matches**	

Case	Formula applied	Computed	NPL's sheet	Result	Note
Alpha Plus (reseller): EUR 2,939 target, market 1 ÷ 0.8712, fee 1.5 %	target_out; 2,939 × 1.015 × 1.147842056933	3,424.11	3,424.11	**matches**	rounded to USDT 3,424
Fair (reseller): EUR 1,079 target, market 1 ÷ 0.8642, fee 0.75 %	target_out; 1,079 × 1.0075 × 1.157139551030	1,257.92	1,257.92	**matches**	rounded to USDT 1,258
D28 rate direction: Ali quotes 1.1738 (USDT per EUR) on a USDT → EUR pair	normalised rate = 1 ÷ 1.1738 at the pair's precision (6)	0.851934	0.851934	**matches**	a direction slip freezes the pair instead of pricing a deal

16 of 17 match; the one marked "sheet differs" is the receiver line of NPL's sheet, which compounds the receiver's part on the gross out; the confirmed rule (§3) puts it on the principal.

20 · Acceptance

1. The engine reproduces every vector of §18 and §19 exactly (to the cent, or to the unit where the sheet rounds), and every posting amount of the Ledger Posting Design v5.9.
2. Fee parts are computed on the principal (§3) and never compounded; a target-out quote applies the fee as a markup on the target (§6).
3. Every rounding step uses the pair's configured option and records the signed residual with its owner (§9).
4. Variances are posted to the account of their cause (§12); the forex true-up of each instalment is posted per line (§14); an earnings offset leaves every client entitlement unchanged (§13).
5. NPL_PROFIT returns remittance earnings, reseller earnings and their sum, per currency, and the same figures through REPORTING_VALUE (§15, §17).

Draft v5.9 — for review with NPL. Vectors A are worked examples; vectors B are NPL's figures from the fee table of 8 October 2026.